



End Term (Even) Semester Regular/Back Examination May 2026

Roll no.....

Name of the Program and semester: B.Tech Core/AI-ML/AI-DS/CS VI SEM

Name of the Course : Full Stack Web Development

Course Code: TCS 693

Time: 3 hours

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question is 20 (twenty).
- (iv) Each sub-question carries 10 marks.

Q1.

(2X10=20 Marks)

- a) Explain in brief what is meant by the term Semantic Elements in HTML. Why are they important, and how do they contribute to the structure and accessibility of a webpage? CO1
- b) Describe how CSS frameworks (like Bootstrap) help in web development. Explain the importance of **responsive design** and how CSS supports it. CO1
- c) Describe the CSS box model in detail. Also, explain pseudo-classes and pseudo-elements in CSS, demonstrating the usage of :hover, :nth-child(), ::before, and ::after. CO1

Q2.

(2X10=20 Marks)

- a) Define DTD (Document Type definition). What is a PCDATA and CDATA section in XML? With the help of XML, create a database of employees and validate it with the help of DTD. CO2
- b) What is the difference between client-side and server-side form validation? Write a JavaScript program that validates a form with fields for Name, Email, Phone Number, and Password, displaying error messages for invalid inputs using regular expression. CO1, CO2
- c) What are JavaScript events? Explain different types of events such as click, mouseover, keypress, and focus. Write a program that Change Color Randomly on Mouse Hover over a rectangular region. CO2

Q3.

(2X10=20 Marks)

- a) Explain difference between GET and a POST method in form handling. Also explain different methods of date object. Display the current date and check whether today is a weekend or weekday. CO2
- b) Explain how jQuery simplifies DOM manipulation and event handling. Write a jQuery script to create a simple slideshow that automatically cycles through images stored in an array. CO2
- c) Explain the concept of backend data loading and also, write a program to retrieve JSON data using AJAX. CO3

Q4.

(2X10=20 Marks)

- a) Design an HTML form to accept the following student details: Student ID, Student Name, Course, Year, and Email. Write a PHP program to connect to a MySQL database and insert the submitted



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data into an existing table named students. Display an appropriate success message if the record is inserted successfully; otherwise, display an error message. CO4

- b) Explain the concept of cookies and session in web development. Write a PHP script to Set a cookie named "user_pref" with an expiry time of 1 day. Also retrieve and display the stored preference. CO4
 - c) What is file handling in PHP? Explain different file modes (read, write, append). Describe how data can be read from an existing file and displayed dynamically on a webpage CO4, CO5
- Q5. (2X10=20 Marks)
- a) Define the MEAN Stack. Explain each of its components—MongoDB, Express.js, Angular, and Node.js in detail. Describe how CRUD operations are performed in a MEAN Stack application. CO5
 - b) Define React.js and explain its core principles. Describe component-based architecture and Virtual DOM in detail. CO5
 - c) What is a Content Management System (CMS)? Describe the key features of a CMS, such as content creation, editing, publishing, and user management. CO5